In this chapter, let us see a few experiments of Faraday, the results and the phenomenon of Electromagnetic Induction. Before that, we will recollect the concept of magnetic flux linked with a surface area.

An anecdote!

Michael Faraday was enormously popular for his lectures as well. In one of his lectures, he demonstrated his experiments which led to the discovery of electromagnetic induction.

At the end of the lecture, one member of the audience approached Faraday and said, "Mr. Faraday, the behaviour of the magnet and the coil of wire was interesting, but what is the use of it?" Faraday answered politely, "Sir, what is the use of a newborn baby?"

Note: We will soon see the greatness of 'that little child' who has now grown as an adult to cater to the energy needs.

4.1.2 Magnetic Flux (Φ_B)

The magnetic flux Φ_B through an area A in a magnetic field is defined as the number of magnetic field lines passing through that area normally and is given by the equation (Figure 4.1(a)).

$$\Phi_B = \int_A \vec{B}.\, d\vec{A} \qquad (4.1)$$

where the integral is taken over the area A and θ is the angle between the direction of the magnetic field and the outward normal to the area.

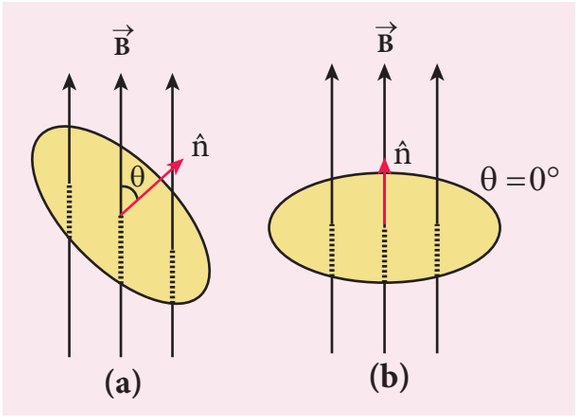


Figure 4.1 Magnetic flux

If the magnetic field $\vec{B}$ is uniform over the area A and is perpendicular to the area as shown in Figure 4.1(b), then the above equation becomes

$$\Phi_B = \int_A \vec{B}.\, d\vec{A} = BA\cos\theta$$
$$= BA \qquad \text{since } \theta = 0°,\ \cos 0° = 1$$

The SI unit of magnetic flux is T m^2. It is also measured in weber or Wb.

1 Wb = 1 T m^2

EXAMPLE 4.1

A circular antenna of area 3 m^2 is installed at a place in Madurai. The plane of the area of antenna is inclined at 47° with the direction of Earth's magnetic field. If the magnitude of Earth's field at that place is 4.1×10^{-5} T find the magnetic flux linked with the antenna.

Solution

$B = 4.1 \times 10^{-5}$ T; $\theta = 90° - 47° = 43°$; $A = 3\text{m}^2$

12th Physics_Vol-1_Unit 4_English Medium.indd 196 23-12-2022 17:56:29